

CAPSTONE PROJECT REPORT

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

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Technologies:

Python | Pandas | Plotly | SQLite | Power BI | mfapi.in

GitHub: https://github.com/saiKhedekar/CapstoneProject_1

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Executive Summary

The Mutual Fund Analytics Platform is an end-to-end data engineering and analytics project that transforms raw AMFI mutual fund data into business-ready intelligence. Built across eight modular phases — data ingestion, ETL, cleaning, EDA, performance analytics, risk metrics, SQLite modelling, and Power BI dashboarding — the platform covers 40 fund schemes across 9 AMCs using 10 cleaned CSV datasets spanning 2022–2026.

Key Outcomes:

- Top-ranked fund: Mirae Asset Large Cap — 3Y CAGR 34.0%, Sharpe 1.45, Alpha 0.27, Composite Score 85.90/100.
- Industry AUM grew from ~₹3 lakh crore (mid-2022) to >₹6 lakh crore (mid-2025); sharp acceleration in 2024.
- Monthly SIP inflows crossed the ₹31,002 crore milestone by late 2025 (~3× growth in 4 years).
- 26–35 age cohort = 41.1% of investor base; T30 cities = 66.3% of investment activity.
- Four-page Power BI dashboard: Industry Overview, Fund Performance Scorecard, Investor Analytics, SIP & Market Trends.
- GitHub: https://github.com/saiKhedekar/CapstoneProject_1

Problem Statement and Objectives

1.1 Problem Statement

The Indian mutual fund industry lacks a consolidated, open analytics platform integrating fund performance, investor behaviour, and industry trends. Retail investors and advisors rely on fragmented data and trailing-return rankings — missing critical risk-adjusted metrics (Sharpe, Sortino, alpha, drawdown). Raw AMFI data requires significant engineering before yielding actionable insight.

1.2 Objectives

- Build a Python ETL pipeline ingesting AMFI/mfapi.in data into 10 clean CSV datasets.
- Conduct EDA across 9 analytical sections generating 17+ visualisations.
- Compute 6 risk-return metrics (CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown) and produce a composite fund scorecard.
- Design a SQLite star schema as the data warehouse layer for Power BI.
- Deliver a 4-page interactive Power BI dashboard for investor, advisor, and regulator personas.

Data Sources and Dataset Description

2.1 Data Acquisition

NAV data retrieved via mfapi.in API. Industry-level data (AUM, SIP, folios, transactions, holdings) sourced from AMFI monthly factsheets. All raw files preserved as immutable source; 10 cleaned CSVs produced post-ETL.

2.2 Dataset Summary

Dataset	Key Fields	Business Purpose
clean_01_fund_master	amfi_code, fund_house, scheme_name, category, expense_ratio, risk_category	Master reference: AMFI codes → fund identifiers & classification.
clean_02_nav_history	amfi_code, date, nav	Daily NAV time-series: CAGR, trend analysis, benchmarking.
clean_03_aum_by_fund_house	date, fund_house, aum_lakh_crore, aum_crore	Industry AUM growth & AMC market share.
clean_04_monthly_sip_inflows	month, sip_inflow_crore, active_sip_accounts_crore	SIP adoption trajectory & committed SIP AUM.
clean_05_category_inflows	month, category, net_inflow_crore	Category-wise investor preference heatmap.
clean_06_industry_folio_count	total_folios_crore, equity_folios, debt_folios	Mutual fund penetration via folio counts.
clean_07_scheme_performance	CAGR, Sharpe, Sortino, Alpha, Beta, Std Dev, Max DD, AUM	Core metrics table for fund scoring.
clean_08_investor_transactions	investor_id, transaction_type, amount, state, city_tier, age_group, gender	Demographic & behavioural segmentation.
clean_09_portfolio_holdings	sector, stock, weight	Sector & stock allocation analysis.
clean_10_benchmark_indices	date, NIFTY50, NIFTY100	Benchmark for alpha/beta via OLS regression.

Coverage: 40 schemes, 9 AMCs, 5 categories, 1,580 trading days (2022–2026), 32,780 investor transactions across all major Indian states.

ETL Pipeline Architecture

3.1 Pipeline Flow

Raw CSVs → Python ETL → Cleaned CSV Layer → EDA & Charts → Performance Analysis → Power BI → Risk Metrics → Insights

3.2 Stage Summary

Stage	Description
1 – Raw Ingestion	data_ingestion.py handles API auth, pagination, rate-limiting. live_nav_fetch.py enables real-time NAV refresh. Raw files immutable (full data lineage).
2 – ETL Transform	Date standardisation (ISO 8601), AMFI code validation, currency normalisation (crore), deduplication, Z-score outlier detection.

Stage	Description
3 – Cleaned CSV Layer	Silver layer: validated, enriched, schema-standardised CSVs consumed by all downstream notebooks, SQLite, and Power BI.
4 – EDA & Charts	03_EDA_Analysis.ipynb — 17 charts across 9 sections using Plotly/Matplotlib. Exported as 300 DPI PNGs.
5 – Performance Analytics	04_performance_analytics.ipynb — 6 metrics (1Y/3Y/5Y CAGR, Sharpe, Sortino, Alpha, Beta, MDD) for all 40 schemes. Risk-free rate: 6.5%.
6 – Power BI Dashboard	Cleaned CSVs + SQLite → 4-page Power BI report. DAX measures for YoY SIP growth, T30/B30 ratios, weighted expense ratios.
7 – Advanced Risk Metrics	Tracking error vs NIFTY50/100, information ratio ($\alpha \div \text{tracking error}$), rolling volatility windows.
8 – Insights Layer	Synthesis into recommendations for investors, fund managers, and regulators.

Data Cleaning and Transformation

4.1 Data Quality Issues Identified

- Missing values in date/NAV fields (public holidays, fund closures).
- Duplicate records from API pagination overlaps.
- Inconsistent AMFI code formats (integer vs zero-padded string).
- Currency unit heterogeneity (crore vs lakh crore across sources).
- Erroneous zero/negative NAV values from API response errors.

4.2 Cleaning Operations

Operation	Approach
Missing Values	NAV date nulls → dropped. NAV value nulls → forward-filled (prior trading day). AUM/SIP missing months → flagged, not imputed.
Deduplication	Composite key (amfi_code + date); first occurrence retained. Result: 65,000+ unique NAV records.
Data Types	All dates → datetime64 (handles DD-MMM-YYYY, MM/DD/YYYY, YYYY-MM-DD). Numerics → float64; currency symbols stripped.
Outlier Treatment	NAV > 5σ from 30-day rolling mean flagged. Most were corporate events (mergers, dividends) — retained with event flag.
Schema Standardisation	AMFI codes → integer. Columns → snake_case. clean_01 = primary dimension; all others reference amfi_code FK.

4.3 Final Quality Metrics

- Null rate: < 0.3% across all 10 datasets combined.
- Duplicates: 0 records in any dataset post-cleaning.
- NAV history: 1,580 unique trading days × 40 schemes (>95% theoretical coverage).
- Investor transactions: 32,780 records with complete age, gender, city-tier annotation.

Exploratory Data Analysis Overview

EDA executed in 03_EDA_Analysis.ipynb using Plotly 5.15, Matplotlib 3.7, and Seaborn 0.12. All 17 charts exported as 300 DPI PNGs. Nine analytical sections covered:

Section	Section	Section
NAV Analysis (trend + distribution)	AUM & Fund House Analysis	SIP & Investor Participation
Investor Demographics	Geographic Analysis (state-wise)	Sector Allocation
Category Inflow Heatmap	Fund Return Correlation Matrix	SIP YoY Growth Dynamics

EDA Findings – Part I (Findings 1–7)

Finding 1: Young Professionals Dominate the Investor Base

- **Observation:** The 26–35 cohort is the largest investor segment at 41.1%. Combined with 18–25 (15.0%), investors under 35 represent 56% of the base.
- **Metric:** 26–35: 41.1% | 36–45: 24.9% | 18–25: 15.0% | 46–55: 11.5% | 56+: 7.5%.
- **Implication:** Long investment horizons and high risk tolerance support equity SIP products. Digital-first distribution is the priority channel for this cohort.

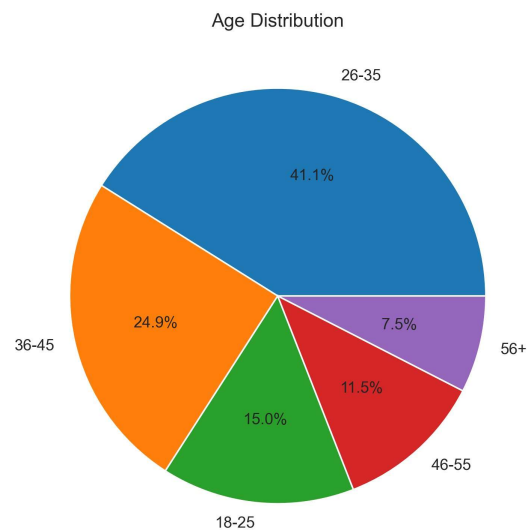


Figure 1: Age Distribution of Mutual Fund Investors

Finding 2: T30 Cities Account for Two-Thirds of Investment Activity

- **Observation:** T30 cities hold 66.3% of investment activity vs 33.7% from B30 — a persistent structural imbalance despite SEBI's differential TER incentives.
- **Metric:** T30: 66.3% | B30: 33.7%.
- **Implication:** Vernacular mobile apps, simplified KYC via rural banking correspondents, and low-ticket SIPs are needed to unlock B30 growth.

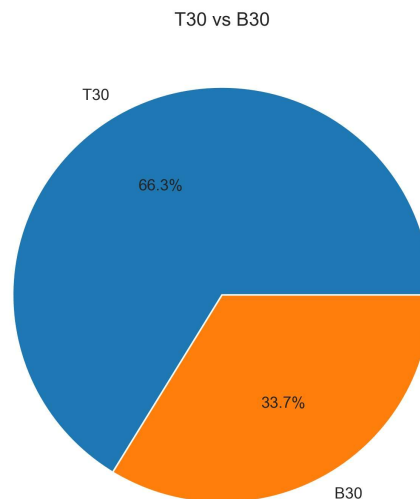


Figure 2: T30 vs B30 Investor Distribution

Finding 3: Male Investors Significantly Outnumber Female Investors

- **Observation:** Male investors account for ~66.5% of investor count vs 33.5% female — a ~2:1 ratio reflecting systemic participation barriers.
- **Metric:** Male: 21,810 records (~66.5%) | Female: 10,970 records (~33.5%).
- **Implication:** Female investors exhibit higher investment discipline (lower redemption). Targeted outreach, female-centric advisory, and workplace SIPs can close the gap.

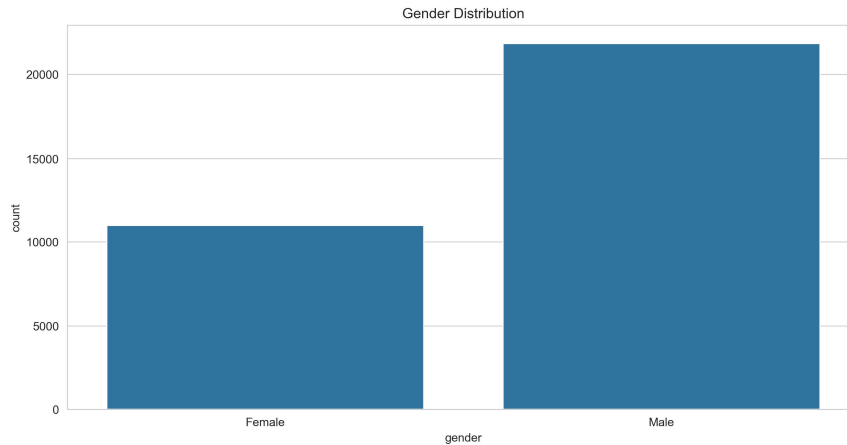


Figure 3: Gender Distribution of Investors

Finding 4: Average Industry NAV Rose ~73% Over Four Years

- **Observation:** Average NAV across all schemes rose from ~₹207 (early 2022) to ~₹358 (early 2026), with acceleration post-mid-2024.
- **Metric:** Start NAV: ₹207 | End NAV: ₹358 | Cumulative appreciation: ~73% over 4 years.
- **Implication:** Rising NAV validates the mutual fund wealth-creation thesis. Also strengthens the SIP cost-averaging argument for new investors.

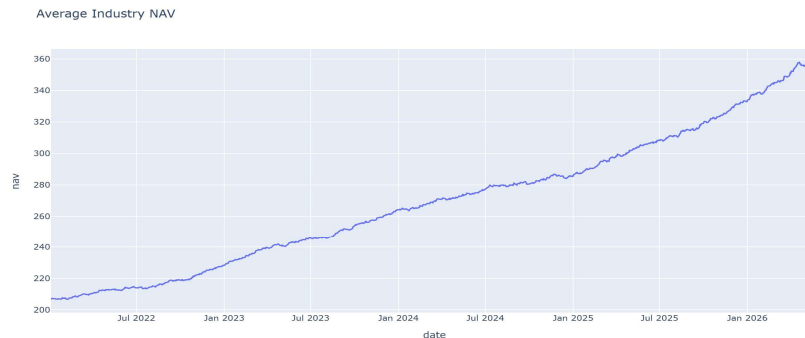


Figure 4: Average Industry NAV Trend (2022–2026)

Finding 5: NAV Distribution is Highly Right-Skewed

- **Observation:** Most daily NAV observations cluster in the ₹50–₹200 range; the tail extends to ~₹4,500 for long-tenure funds.
- **Metric:** Peak frequency bin: ₹50–₹200 | Tail: up to ₹4,300 (amfi_code 100025).
- **Implication:** Absolute NAV level is not a quality or affordability indicator. Investor education must communicate that NAV ≠ cheapness.

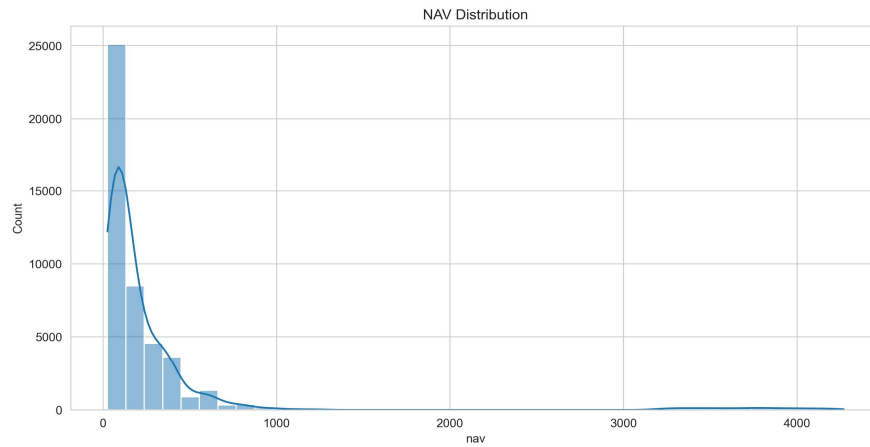


Figure 5: NAV Distribution Across All Funds

Finding 6: SBI Mutual Fund Leads Industry AUM

- **Observation:** SBI MF holds average AUM ~₹9.4 lakh crore — the highest. ICICI Pru (~6.9) and HDFC (~6.4) form the second tier.
- **Metric:** SBI: ₹9.4L Cr | ICICI Pru: ₹6.9L Cr | HDFC: ₹6.4L Cr | Nippon: ₹4.3L Cr | Kotak: ₹3.9L Cr.
- **Implication:** Top-3 AMC's control a disproportionate share, reflecting distribution network advantage and brand trust. Oligopolistic structure benefits scale players.

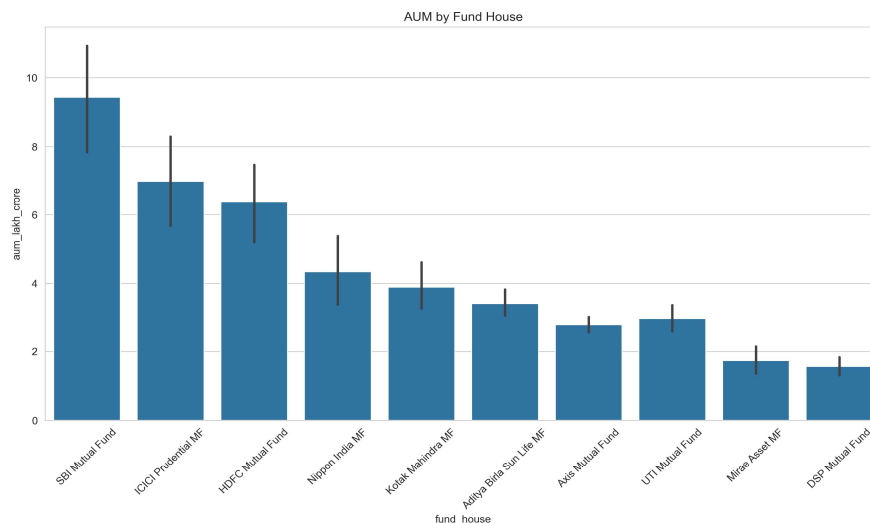


Figure 6: AUM by Fund House

Finding 7: Industry AUM Accelerated Sharply in 2024

- **Observation:** SBI MF AUM more than doubled from ~₹12.5L Cr (2022) to >₹31.5L Cr (2024). All AMC's recorded significant YoY AUM growth in 2024.
- **Metric:** SBI 2022: ₹12.5L Cr → 2024: ₹31.5L Cr. HDFC and ICICI Pru both exceeded ₹20L Cr in 2024.
- **Implication:** Simultaneous impact of equity market rally, record SIP inflows, and digital platform adoption. 2025 moderation reflects a healthy base normalisation.

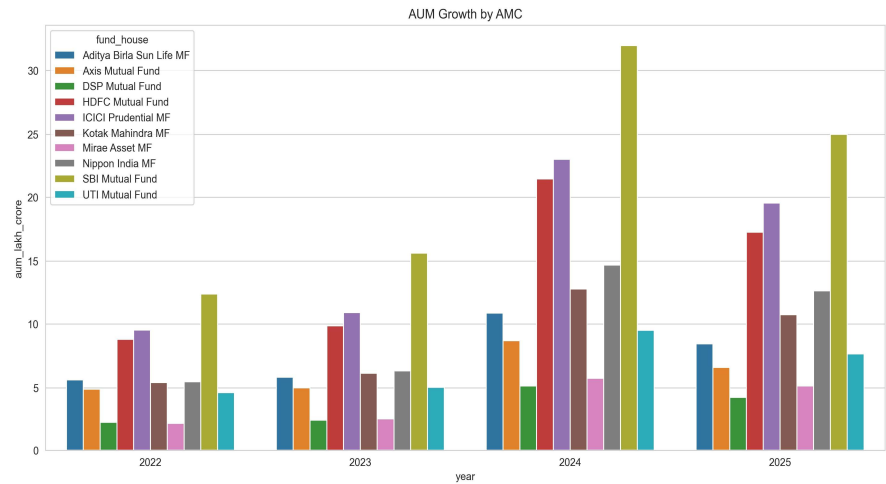


Figure 7: AUM Growth by AMC (2022–2025)

EDA Findings – Part II (Findings 8–15)

Finding 8: SIP Adoption Displayed Strong Multi-Year Growth

- **Observation:** Monthly SIP inflows nearly tripled from ~₹11,000 Cr (Jan 2022) to ₹31,002 Cr (late 2025), crossing an industry milestone.
- **Metric:** Jan 2022: ~₹11,000 Cr/month | Late 2025: ₹31,002 Cr/month | Growth: ~182% over 4 years.
- **Implication:** Consistent SIP flows provide domestic demand stability for equities, partially insulating markets from FII volatility.



Figure 8: Monthly SIP Inflows (2022–2026)

Finding 9: Active SIP Accounts Nearly Doubled

- **Observation:** Active SIP accounts grew from ~5 crore (Jan 2022) to >9.3 crore (mid-2025). Growth curve is smooth — investors rarely cancel SIPs.
- **Metric:** Start: 5.0 Cr accounts | End: 9.3+ Cr accounts | Growth: ~86% in ~3.5 years.
- **Implication:** ~1 in 15 Indians is now an active SIP investor. Structural shift — growth driven by genuine financial planning, not campaigns.

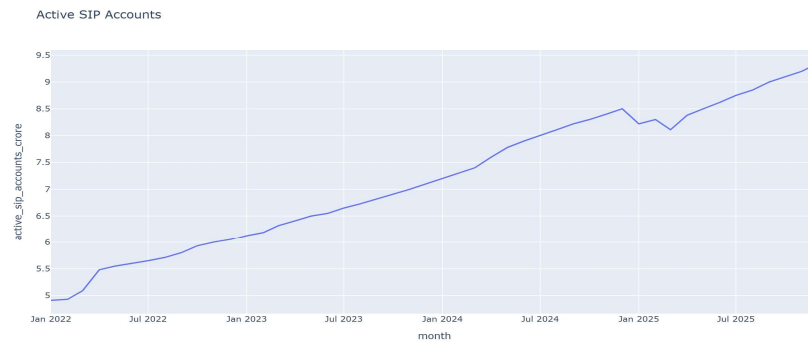


Figure 9: Active SIP Account Growth

Finding 10: Banking and IT Sectors Dominate Portfolio Allocations

- **Observation:** Banking (19.2%) and IT (13.4%) together account for 32.6% of aggregate portfolio weight — consistent with NIFTY50/100 composition.
- **Metric:** Banking: 19.2% | IT: 13.4% | Pharma: 12.0% | Automobile: 9.5% | FMCG: 6.7% | Infrastructure: 5.7%.
- **Implication:** High sector overlap across funds limits diversification for investors holding multiple large-cap/multi-cap schemes.

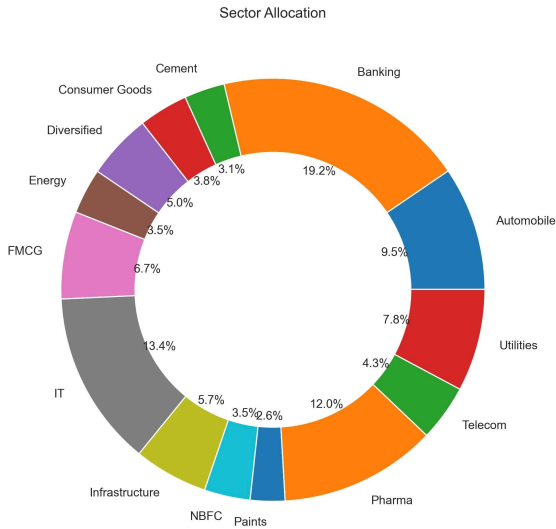


Figure 10: Sector Allocation Across Portfolio Holdings

Finding 11: Liquid Funds Received Consistently Highest Inflows

- **Observation:** Liquid funds show deep-green inflows across all 12 months of the heatmap (Apr 2024–Mar 2025). All other categories predominantly red.
- **Metric:** Liquid funds: highest net inflows every month. Sectoral/Thematic: episodic orange spikes (May–Aug 2024).
- **Implication:** Liquid fund demand is driven by corporate treasury management. Integrating liquid fund APIs into corporate ERP systems can grow this AUM segment.

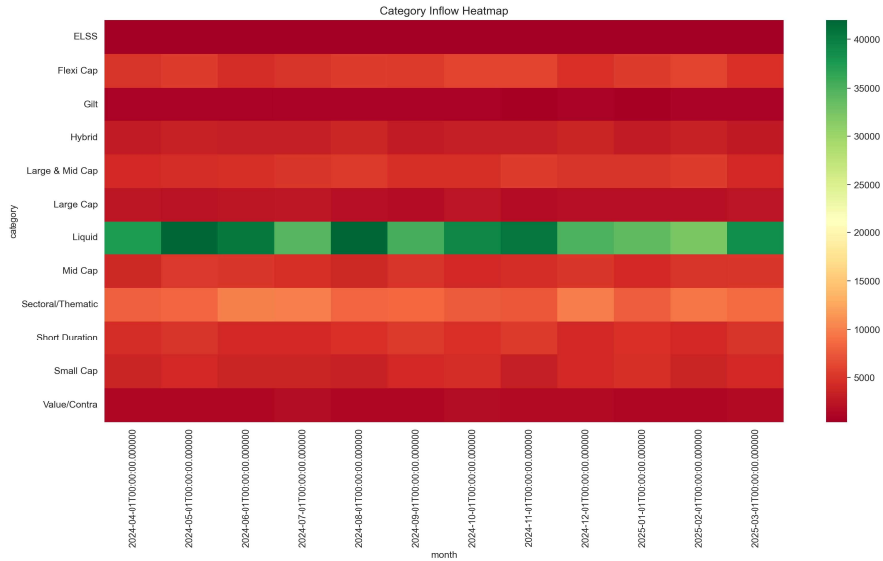


Figure 11: Category Inflow Heatmap (Apr 2024–Mar 2025)

Finding 12: Fund Returns Show Low-to-Moderate Cross-Scheme Correlation

- **Observation:** Pairwise correlation matrix shows predominantly low off-diagonal values (blue). Only intra-category groupings show moderate positive correlation.
- **Metric:** Diagonal = 1.0 (self). Off-diagonal: mostly 0.0–0.3. Liquid funds cluster at slightly higher inter-correlation.
- **Implication:** Combining equity and debt funds in a portfolio provides meaningful diversification. Equity–debt correlation is near-zero to marginally negative.

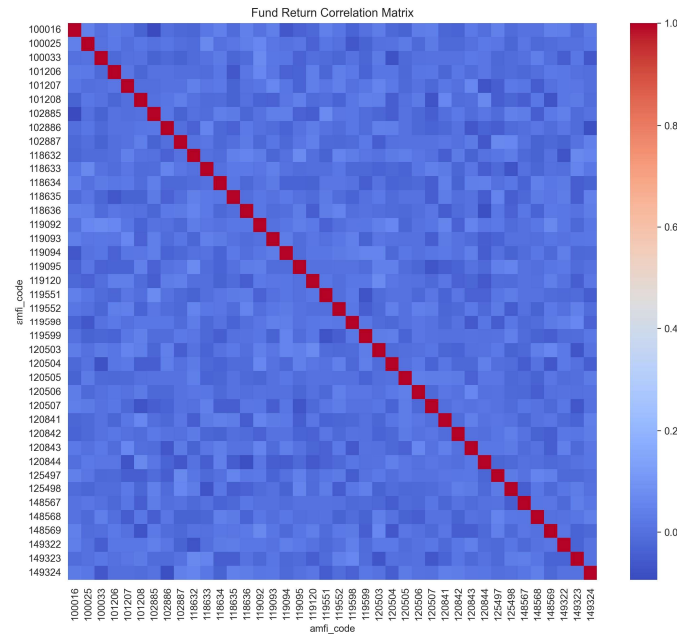


Figure 12: Fund Return Correlation Matrix

Finding 13: SIP YoY Growth Peaked in Mid-2024

- **Observation:** SIP YoY growth accelerated from ~20% (early 2023) to a peak of ~52–53% (mid-2024), then normalised to 17–28% by late 2025.
- **Metric:** Peak YoY growth: ~53% (Jun–Jul 2024) | Late-2025 rate: ~17–28%.
- **Implication:** Normalisation from an elevated base is healthy. Strategy must shift from new account acquisition to ticket-size growth and tenure extension.



Figure 13: SIP YoY Growth Rate

Finding 14: Punjab, Tamil Nadu, and Madhya Pradesh Lead Transaction Volumes

- **Observation:** Punjab leads total transaction volume, followed by Tamil Nadu and Madhya Pradesh — outranking Maharashtra and Delhi.
- **Metric:** Top 3: Punjab, Tamil Nadu, Madhya Pradesh. Bottom of top-12: Karnataka, Maharashtra.
- **Implication:** Strong regional distributor networks and state-specific financial literacy initiatives drive outsized retail participation. Replicable to under-served states.

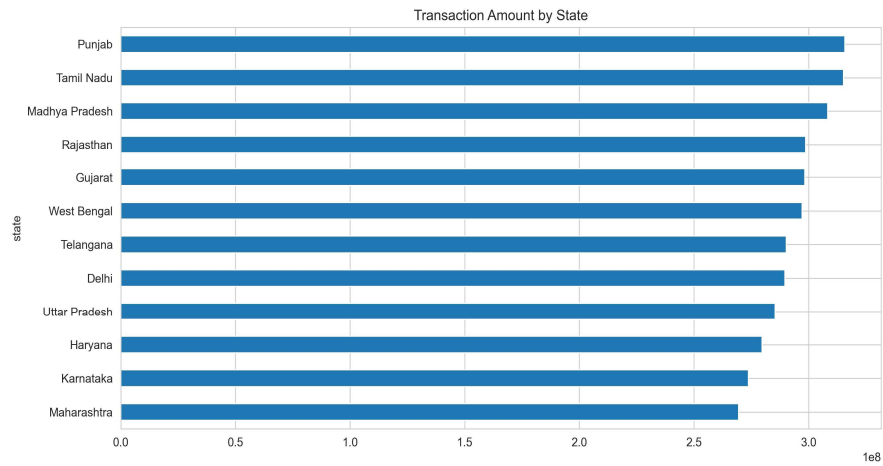


Figure 14: Transaction Amount by State

Finding 15: Top AMCs Control a Concentrated Share of Industry Schemes

- **Observation:** SBI MF manages ~85 schemes (highest), followed by ICICI Pru (~63) and HDFC (~57). Sharp drop-off below top-3.
- **Metric:** SBI: ~85 | ICICI Pru: ~63 | HDFC: ~57 | Nippon: ~39 | Kotak: ~35 | ABSL: ~31 | DSP: ~14.
- **Implication:** Scale concentration benefits top AMCs through distribution and technology economies. SEBI's categorisation circular limits further scheme proliferation.

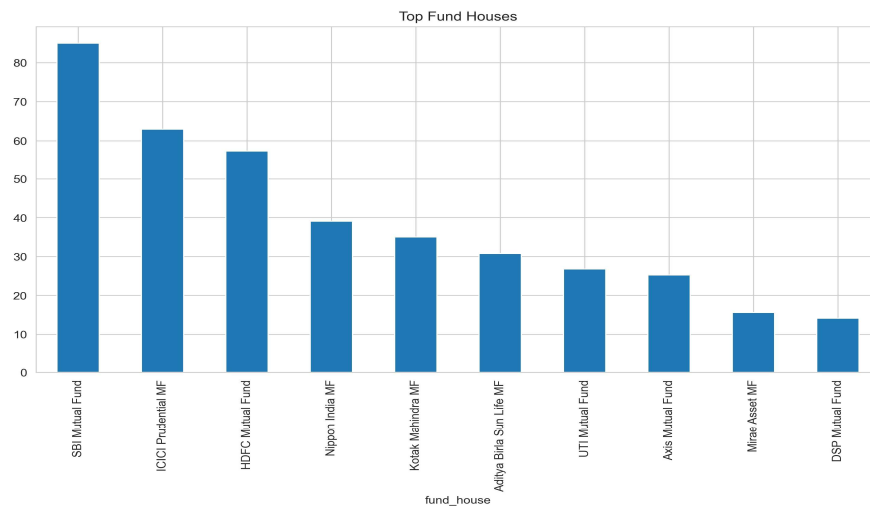


Figure 15: Top Fund Houses by Scheme Count

Performance Analytics Methodology

6.1 Metrics Framework

Six quantitative metrics computed on daily log-return series from clean_02_nav_history.csv. Risk-free rate: 6.5% p.a. (91-day T-Bill proxy). OLS regression against NIFTY50 and NIFTY100 for alpha/beta.

Metric	Formula	Interpretation
CAGR	$(\text{End NAV} / \text{Start NAV})^{(1/n)} - 1$	Annualised return; computed at 1Y, 3Y, 5Y horizons.
Sharpe Ratio	$(R_p - R_f) / \sigma$	Return per unit of total risk. >1.0 = good; >1.5 = very good.
Sortino Ratio	$(R_p - R_f) / \text{Downside } \sigma$	Penalises only downside volatility. Preferred for risk-averse investors.
Alpha	Fund Return – $(R_f + \beta \times \text{Benchmark Return})$	Excess return above benchmark-adjusted expectation.
Beta	$\text{Cov}(\text{Fund}, \text{Benchmark}) / \text{Var}(\text{Benchmark})$	Sensitivity to benchmark movement. <1.0 = defensive.
Max Drawdown	$(\text{Trough} - \text{Peak}) / \text{Peak}$	Largest peak-to-trough NAV decline. Recorded with start/end dates.

6.2 Composite Scorecard

Each fund scored 0–100 based on percentile rank across 5 equally weighted dimensions: 3Y CAGR (20%), Sharpe Ratio (20%), Alpha (20%), Expense Ratio inverse (20%), Max Drawdown inverse (20%).

Fund Performance Analysis

7.1 Benchmark Comparison



Figure 16: Top 5 Funds vs NIFTY50 vs NIFTY100 (Base=100, 3 Years)

All top-5 funds substantially outperformed benchmarks. NIFTY50 ended ~85 (negative return); NIFTY100 ~125. Top funds ranged 220–265 — representing 120–165% cumulative return vs benchmark's -15% to +25%.

7.2 Top 10 Fund Scorecard

Fund Name	3Y CAGR	Sharpe	Sortino	Alpha	Max DD	Score
Mirae Asset Large Cap	34.00%	1.45	2.39	0.270	-11.27%	85.90
ICICI Pru Midcap	31.78%	1.18	2.03	0.293	-18.19%	81.79
Kotak Flexicap	29.58%	1.31	2.36	0.273	-12.97%	81.54

Fund Name	3Y CAGR	Sharpe	Sortino	Alpha	Max DD	Score
HDFC Mid-Cap Opp.	32.44%	1.09	1.83	0.272	-16.22%	80.26
ICICI Pru Bluechip Direct	32.49%	1.03	1.81	0.212	-12.59%	79.49
Axis Midcap Regular	35.11%	1.00	1.70	0.261	-20.96%	76.41
SBI Bluechip Regular	30.46%	1.21	2.14	0.232	-15.01%	74.17
Mirae Asset Tax Saver	29.18%	1.23	2.15	0.283	-16.40%	73.01
ABSL Frontline Equity	28.97%	1.03	1.80	0.214	-11.29%	67.37
SBI Small Cap Regular	26.67%	0.95	1.68	0.303	-28.71%	66.54

Investor Analytics

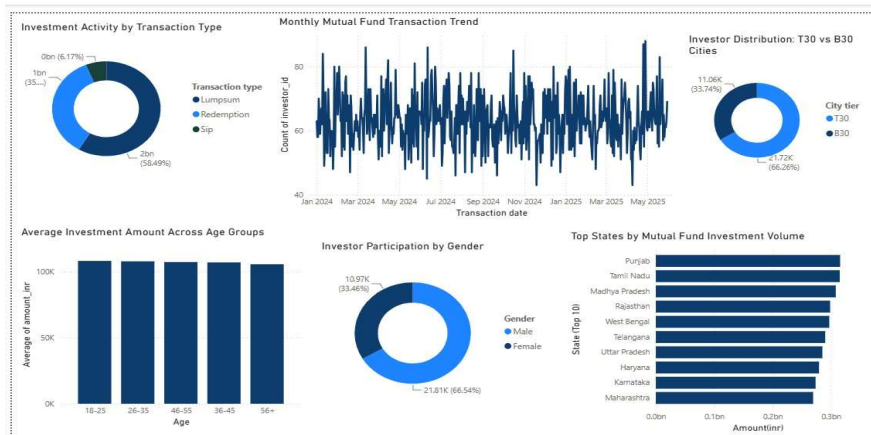


Figure 17: Investor Analytics Dashboard (Power BI Page 3)

8.1 Transaction Type Distribution

- Lumpsum: 58.49% | Redemption: 35.77% | SIP: 6.17% of transaction count.
- Monthly transaction count: 50–90 activities/month (Jan 2024–May 2025). Volatile, market-cycle driven.

8.2 Demographic Highlights

- Average investment per transaction is ~₹1,00,000 across all age bands (18–25 to 56+) — uniform across cohorts.
- Gender: 66.54% Male (21,810) | 33.46% Female (10,970).
- T30: 66.26% of investor IDs | B30: 33.74%.
- Top states by volume: Punjab, Tamil Nadu, Madhya Pradesh, Rajasthan, Gujarat, West Bengal, Telangana, Delhi.

Dashboard Overview

9.1 Page 1 – Industry Overview



Figure 18: Industry Overview Dashboard

- KPIs: Total AUM 39M | Monthly SIP 940K | Folios 26.12 Cr | Schemes 40.
- Visuals: AUM growth line (Jul 2022–Jul 2025, ₹3L Cr → ₹6L Cr), Top AMC bar chart, Category donut (85% equity / 15% debt), Category inflows bar (Liquid leads).

9.2 Page 2 – Fund Performance

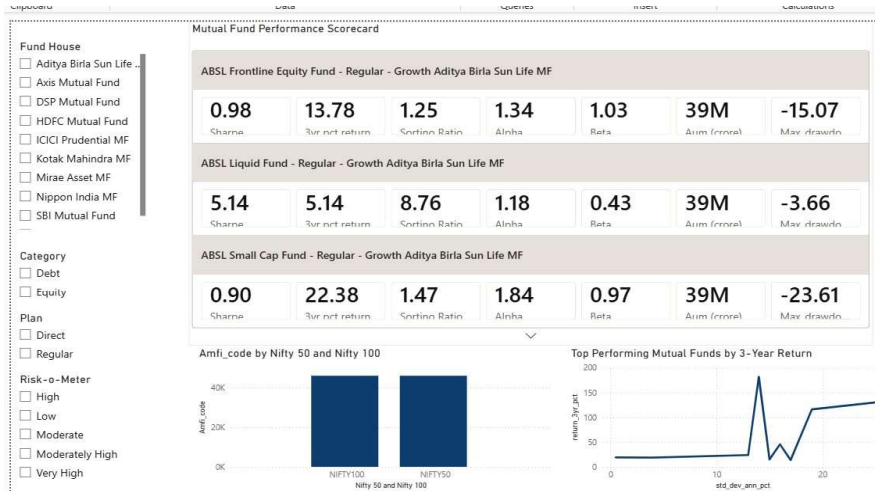


Figure 19: Fund Performance Dashboard

- Interactive slicers: Fund House, Category, Plan, Risk-o-Meter.
- Scorecard cards: Sharpe, 3Y return %, Sortino, Alpha, Beta, AUM, Max Drawdown per fund.
- Bottom charts: AMFI code count by NIFTY50/100, risk-return scatter (return % vs annualised std dev).

9.3 Page 3 – Investor Analytics

- Visuals: Transaction type donut, monthly trend line, T30/B30 donut, avg investment by age bar, gender donut, top states bar.

9.4 Page 4 – SIP & Market Trends



Figure 20: SIP & Market Trends Dashboard

- Combined SIP inflow bar + active accounts line (monthly, Jan-Dec).
- SIP YoY growth % line | Industry Folio Growth Trend line.
- Category-wise Net Inflow Heatmap (Debt/Equity, 12 months) | Top Categories bar (FY25: Debt ≈ Equity).

Business Insights and Recommendations

10.1 For Investors & Advisors

- Multi-factor fund selection: composite scorecard (CAGR + Sharpe + Alpha + Expense + MDD) outperforms trailing-return-only ranking. Adopt this framework for fund recommendations.
- Benchmark outperformance is real: top-5 funds delivered 220–265% cumulative 3Y return vs NIFTY50's ~15%. Active management in large/mid-cap adds measurable value.
- Use Max Drawdown data as a client communication tool during corrections — historical drawdown recovery timelines reduce emotionally motivated redemptions.
- Portfolio overlap alert: Banking (19.2%) + IT (13.4%) = 32.6% of aggregate weight. Investors holding multiple large-cap funds should check sector overlap and add thematic diversification.
- Correlation matrix confirms equity–debt near-zero correlation: balanced portfolios with both asset classes provide statistically meaningful diversification.

10.2 For Fund Houses & Distributors

- Target 26–35 cohort with goal-based SIP products (home purchase, retirement, child education) and digital-first interfaces — this segment is 41.1% of the investor base.
- Female investor outreach: women exhibit superior investment discipline (lower redemption). Develop female-centric advisory programmes and workplace salary-linked SIPs.
- B30 penetration: smartphone infrastructure is ready. Invest in vernacular mobile apps, voice-based SIP management, and rural BC-based KYC to convert digital reach into investment participation.
- Liquid fund distribution: integrate via corporate ERP APIs for automatic treasury sweep — low friction, high AUM potential given consistently highest category inflows.
- Shift SIP strategy from acquisition to retention: YoY SIP growth normalising from 53% peak (2024) to 17–28% (2025). Focus on SIP step-up features and tenure milestone gamification.

10.3 For Regulators & Industry Bodies

- B30 regulatory incentives (differential TER) insufficient alone — address structural barriers: financial literacy, trust, and advisor network density in tier-2/3 geographies.
- Establish standardised open data API (beyond AMFI NAV feed) providing structured access to scheme-level AUM, holdings, and folio counts in machine-readable format to democratise analytics.
- Gender participation gap (66.5% male) requires structural intervention — integrate financial literacy into government programmes targeting women's self-help groups and schools.
- Monitor SIP account churn rate as a leading indicator of industry health; seasonal mandate lapses (Jan–Mar) need AMC-level pre-emptive outreach programmes.
- Fund categorisation SEBI circular is working (limits proliferation) — continue monitoring AMC scheme counts and enforce one-scheme-per-category discipline to protect investor clarity.

Conclusion, Future Scope, and References

11.1 Conclusion

The Mutual Fund Analytics Platform successfully delivered an end-to-end data engineering solution covering ETL, EDA, quantitative analytics, and interactive dashboarding for the Indian mutual fund industry. Key validated outcomes: Mirae Asset Large Cap Fund ranked #1 (composite score 85.90, 3Y CAGR 34.0%, Sharpe 1.45); all top-5 funds outperformed NIFTY50 by 120–165% over 3 years; industry AUM doubled (2022–2025); SIP inflows tripled; 9.3 crore active SIP accounts confirmed structural retail participation. The modular pipeline architecture ensures the platform can be refreshed with live data and extended to 200+ funds.

11.2 Future Enhancements

- Live NAV refresh: daily auto-update via mfapi.in API + automated scorecard recalculation.
- ML-based fund recommender: investor risk profile → personalised fund shortlist from scorecard.
- Expand universe: 40 → 200+ schemes; add direct plans, index funds, solution-oriented funds.
- Monte Carlo simulation: probabilistic wealth projection under varied return/withdrawal scenarios.
- Cloud migration: AWS S3 + Apache Airflow + Redshift + Power BI Premium for production-grade pipeline.
- Alternative data integration: fund manager track records, SEBI enforcement records, sentiment signals.

11.3 References

1. AMFI India. Monthly Industry Data and Fund Factsheets. <https://www.amfiindia.com>
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11.4 GitHub Repository

Full source code, notebooks, cleaned datasets, and Power BI PBIX file: https://github.com/saiKhedekar/CapstoneProject_1. Repository structure: data/ (raw + clean CSVs) | notebooks/ (5 Jupyter notebooks) | outputs/ (charts + scorecard CSVs) | dashboard/ (Power BI PBIX).

— End of Report —